

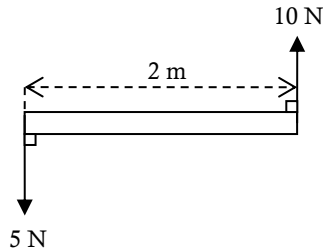
INTRODUCTION TO MOMENTS

12/03/26

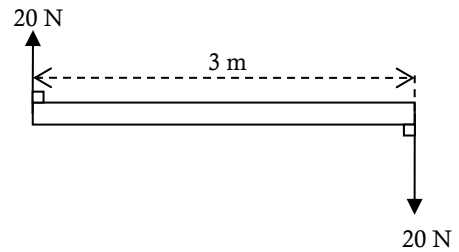
1. In each of the following cases, find the resultant moment about:

- the left end of the bar;
- the centre of the bar;
- the right end of the bar.

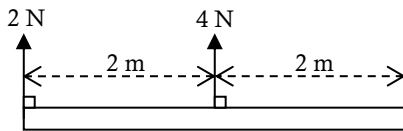
a.



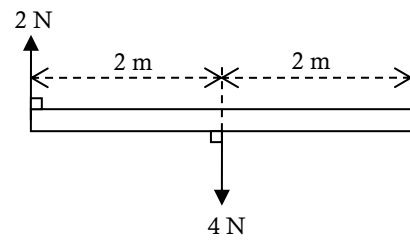
b.



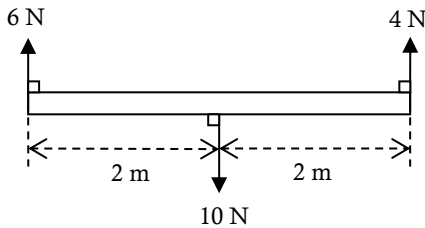
c.



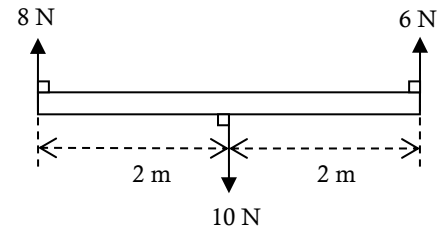
d.



e.

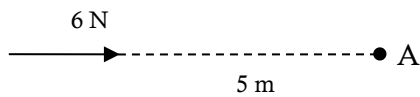


f.

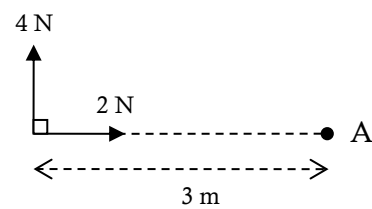


2. Find the resultant moment about the point A of the systems of forces shown.

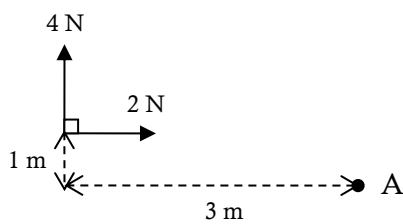
a.



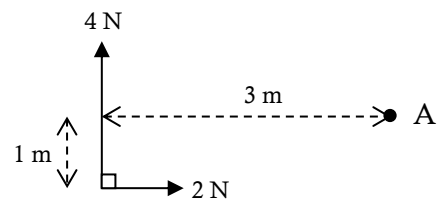
b.



c.

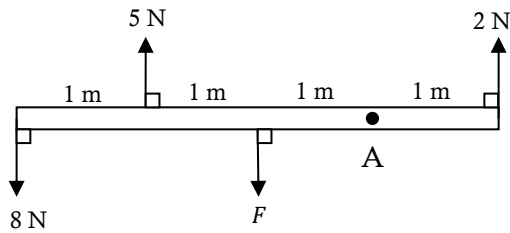


d.

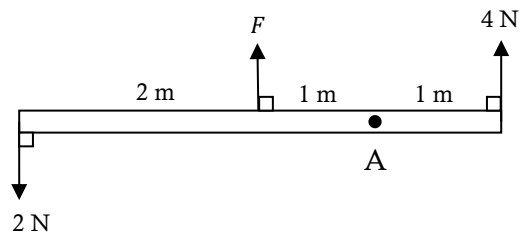


3. In each case, find the magnitude of the force F which will make the moment about the point A equal to 10 Nm clockwise:

a.



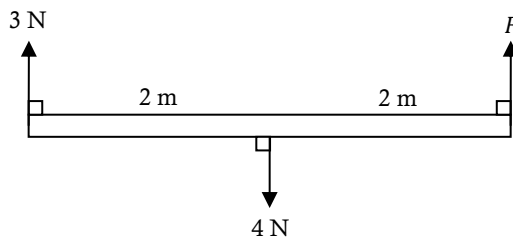
b.



4. Find the distance from the left hand edge of the rod of the point about which the moment is of the system of forces is 0 Nm, when:

a. $F = 5 \text{ N}$

b. $F = 9 \text{ N}$

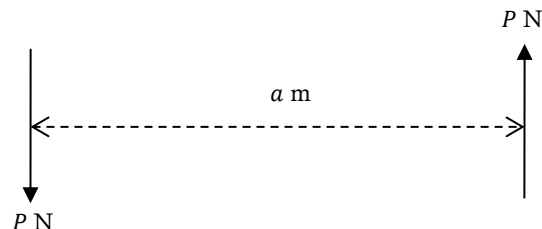


What happens if $F = 1$?

Bonus questions

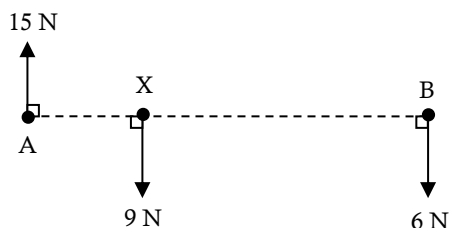
5. Two forces of equal magnitude, opposite directions, and not acting along the same line are said to form a **couple** (see below, for example). A couple has a turning effect, but cannot produce a translatory effect.

Prove that the magnitude of the moment of the couple shown is $aP \text{ Nm}$ about **every point**.

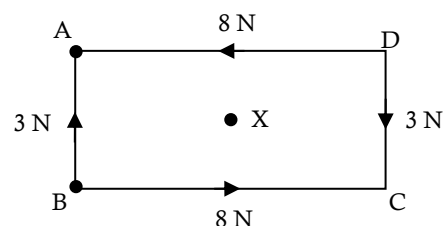


6. Any system of forces which has no translatory effect is either in equilibrium, or reduces to a couple. Verify this in the following cases by calculating, in terms of x and y , the moments of the following systems of forces about the points A, B, and X.

a. $AB = 10$ and $AX = x$



b. $AB = 2x$, $BC = 2y$, and X is the centre of ABCD.



In each case, identify any values of x and y for which the system is actually in equilibrium, and not, therefore, a couple.